

Notes: Dembo, Kariv, Polisson & Quah (JPE, 2026)

Ever since Allais

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1 Big picture

- **Question.** Which axioms of expected utility theory actually fail in rich individual choice data under risk: independence, ordering, or monotonicity with respect to first-order stochastic dominance?
- **Motivation.** The classic Allais critique focuses on violations of the independence axiom. Most non-EUT models relax independence but retain ordering and monotonicity. The paper asks whether this focus is empirically justified.
- **Experimental innovation.** Subjects choose allocations of three state-contingent securities from three-dimensional budget sets using a graphical point-and-click interface.
- **Why three dimensions matter.** With two goods, WARP and GARP are observationally equivalent. With three goods, the experiment can distinguish pairwise inconsistency, cyclical inconsistency, monotonicity violations, and violations of expected utility.
- **Revealed-preference strategy.** The authors use individual-level nonparametric tests rather than fitting one parametric utility model to pooled data.
- **Three rationalizability scores.** The paper computes e^* for basic rationalizability, e^{**} for FOSD-rationalizability, and e^{***} for EUT-rationalizability.
- **Main empirical result.** Most departures from EUT come from failures of ordering or FOSD monotonicity, not from failures of independence alone.
- **Key quantitative fact.** At threshold 0.95, 63.1% of subjects are rationalizable, 28.0% are FOSD-rationalizable, and 16.1% are EUT-rationalizable.
- **Difference-in-differences finding.** For 85.1% of subjects, the gap between perfect rationalizability and FOSD-rationalizability exceeds the gap between FOSD-rationalizability and EUT-rationalizability.
- **Interpretation.** If subjects often violate WARP, GARP, or FOSD, then models that merely relax independence while maintaining the other axioms cannot explain most observed behavior.
- **Takeaway.** The Allais critique is incomplete: the empirical failure of EUT appears to lie less in independence alone and more in deeper violations of ordering and monotonicity.

2 Incremental contribution of the paper

2.1 Contribution relative to Allais-style experiments

- Traditional Allais experiments test independence using a small number of paired lotteries.
- Those experiments typically assume, rather than test, completeness, transitivity, and FOSD monotonicity.
- This paper designs an experiment that can test the full set of axioms underlying EUT.
- The key innovation is replacing a small set of pairwise choices with many budget choices over three state-contingent securities.

Interpretation: The incremental contribution is to move from testing one famous implication of EUT to decomposing the sources of EUT failure axiom by axiom.

2.2 Contribution relative to budget-choice revealed-preference experiments

- Earlier budget-choice experiments often use two-dimensional budget lines.
- With only two goods, WARP and GARP cannot be separated: any GARP violation contains a WARP violation.

- By using three-dimensional budget sets, the paper can distinguish incomplete preferences, nontransitive preferences, and monotonicity violations.
- The design also provides more power to separate EUT from non-EUT models that respect FOSD.

Interpretation: The move from 2D to 3D is not cosmetic. It changes which axioms are identifiable in revealed-preference data.

2.3 Contribution relative to non-EUT modeling

- Prominent non-EUT theories such as rank-dependent utility, weighted expected utility, disappointment aversion, and related models usually retain ordering and FOSD monotonicity.
- The paper shows that many subjects' choices fail exactly those retained axioms.
- Therefore, even successful non-EUT alternatives may explain only a small part of the empirical departure from EUT.

Interpretation: The paper asks a sharper question than “EUT or non-EUT?” It asks whether the empirical failure of EUT is of the kind that leading non-EUT models are designed to repair.

2.4 Contribution relative to revealed-preference econometrics

- The paper builds on Afriat-style revealed preference and on Polisson, Quah, and Renou's nonparametric tests of decision models under risk.
- It applies those tests to newly collected individual-level 3D experimental data.
- It develops a nonparametric statistical comparison of rationalizability-score gaps, especially the difference between $1 - e^{**}$ and $e^{**} - e^{***}$.

Interpretation: The paper contributes methodologically by combining high-dimensional experimental design with individual-level statistical revealed-preference tests.

3 Experimental design and measurement

3.1 State-contingent securities

- Each subject chooses allocations of tokens across three securities.
- There are three equally likely states.
- A unit of security s pays in state s and nothing in the other states.
- Each decision is a choice from a linear budget set in three dimensions.

An observation is a price vector $p_i = (p_{1i}, p_{2i}, p_{3i})$ and a chosen allocation $x_i = (x_{1i}, x_{2i}, x_{3i})$ satisfying

$$p_i \cdot x_i = 1. \tag{1}$$

Interpretation: The allocation is a lottery over monetary payoffs. Choosing securities means choosing how much wealth to allocate to each state of the world.

3.2 Why budget choices are useful

- Budget sets generate many revealed-preference inequalities from each subject.
- They allow nonparametric tests: the researcher does not need to assume a parametric utility function.
- They generate individual-level data rich enough to evaluate multiple axioms separately.

Interpretation: The point of the experimental design is not to estimate a single risk-aversion parameter. It is to test structural restrictions imposed by different theories of choice under risk.

3.3 Basic rationalizability and GARP

A dataset $D = \{(p_i, x_i)\}_{i=1}^T$ is rationalizable if there exists a utility function U such that for every observation i ,

$$x_i \in \arg \max_{x \in B_i} U(x), \quad (2)$$

where

$$B_i = \{x : p_i \cdot x \leq p_i \cdot x_i\}. \quad (3)$$

GARP rules out revealed-preference cycles. If x_i is revealed preferred to x_j through a chain of choices, then x_j cannot be strictly affordable when x_i was chosen.

Interpretation: Afriat's theorem makes GARP the exact finite-data test for maximization of a locally nonsatiated utility function.

3.4 WARP versus GARP

- WARP rules out pairwise revealed-preference reversals.
- GARP rules out longer cycles.
- In two-good settings, every GARP violation contains a WARP violation.
- In three-good settings, GARP can fail without WARP failing.

Interpretation: This is why the paper's three-security design can distinguish incomplete preferences from nontransitive preferences.

3.5 CCEI rationalizability score

Afriat's critical cost efficiency index e^* is the largest number $e \in (0, 1]$ such that revealed preference satisfies GARP after shrinking each budget set by e .

Define the modified revealed-preference relation:

$$x_i R(e) x_j \quad \text{if} \quad e p_i \cdot x_i \geq p_i \cdot x_j. \quad (4)$$

Then e^* is the maximal e for which $R(e)$ satisfies GARP.

Interpretation: If $e^* = 0.95$, then at most a 5% budget reduction is needed to eliminate all revealed-preference violations. Scores closer to one indicate more rationalizable behavior.

3.6 FOSD-rationalizability

A utility function is FOSD-increasing if it assigns higher utility to lotteries that first-order stochastically dominate others. The corresponding score is e^{**} .

$$e^{**} \leq e^*. \quad (5)$$

Interpretation: FOSD violations are often interpreted as mistakes because they involve choosing a lottery that is dominated by another feasible lottery.

3.7 EUT-rationalizability

A dataset is EUT-rationalizable if there exists a Bernoulli utility function v such that choices maximize

$$U(x) = \frac{1}{3} \sum_{s=1}^3 v(x_s). \quad (6)$$

The corresponding CCEI-type score is e^{***} , and it satisfies

$$e^{***} \leq e^{**} \leq e^*. \quad (7)$$

Interpretation: This nesting makes it possible to decompose EUT violations. A gap between e^{**} and e^{***} points to independence-type violations, whereas a gap between e^* and e^{**} points to monotonicity violations.

4 Theoretical and empirical logic

4.1 Probability triangle and non-EUT theories

Read:

- Figure 2 shows the probability simplex for three outcomes.
- Ordering and transitivity guarantee nonintersecting indifference curves.
- FOSD monotonicity restricts the direction in which preferences increase.
- EUT adds independence, implying parallel straight indifference curves.
- Non-EUT models such as weighted expected utility and rank-dependent utility bend or rotate indifference curves but retain ordering and monotonicity.

Interpretation: This visual explains why non-EUT models relax only a subset of EUT. If a subject fails ordering or monotonicity, those models will not explain the behavior.

4.2 Decomposing EUT failures

The three scores separate sources of EUT failure:

$$1 - e^{**} \quad \text{measures violation of basic rationalizability plus FOSD monotonicity,} \quad (8)$$

$$e^{**} - e^{***} \quad \text{measures the incremental loss from imposing EUT after FOSD.} \quad (9)$$

- If $1 - e^{**}$ is large and $e^{**} - e^{***}$ is small, the main failure is not independence.
- If $e^{**} - e^{***}$ is large, then non-EUT models that retain FOSD may explain much of the departure from EUT.

Interpretation: The paper’s headline conclusion follows from comparing these two gaps at the subject level.

4.3 Power comparison between 2D and 3D experiments

The paper performs a Bronars-style simulation. Simulated subjects choose randomly subject to FOSD rationalizability, and the authors compute EUT-rationalizability scores.

Interpretation: The 3D design has more power to detect non-EUT behavior because EUT imposes richer restrictions on choices over three states than over two equally likely states.

5 Main results

5.1 Aggregate score distributions

- Only 16.1% of subjects are perfectly rationalizable ($e^* = 1$).
- No subject is perfectly FOSD-rationalizable or perfectly EUT-rationalizable.
- At threshold 0.95, 63.1% are rationalizable, 28.0% are FOSD-rationalizable, and 16.1% are EUT-rationalizable.

Interpretation: The drop from e^* to e^{**} is much larger than the drop from e^{**} to e^{***} . This is the core empirical fact.

5.2 WARP and GARP violations

- Median WARP violations equal 4.
- Median GARP violations equal 7.
- Most GARP violations contain WARP violations.
- A minority of subjects have GARP violations without WARP violations, which the 3D design can detect.

Interpretation: Failures of basic ordering are present and empirically important. This matters because many models assume ordering before relaxing independence.

5.3 Individual-level scatterplots

- Figure 5 plots e^* against e^{**} and e^{**} against e^{***} .
- Many subjects lie far below the first 45-degree line, showing meaningful FOSD-rationalizability failures.
- Much less dispersion appears between e^{**} and e^{***} for many subjects.

Interpretation: The visual evidence reinforces the decomposition: the main loss often occurs before imposing EUT.

5.4 Difference-in-differences evidence

- Figure 6 compares $1 - e^{**}$ against $e^{**} - e^{***}$.
- 85.1% of subjects fall below the 45-degree line.
- For 74.4% of subjects, the first gap is at least twice as large as the second gap.

- Figure 8 adds statistical testing and shows that many statistically significant cases support the same conclusion.

Interpretation: For the vast majority of subjects, deviations from FOSD rationalizability dominate deviations from FOSD rationalizability to EUT rationalizability.

6 Tables and figures

6.1 Figures 1 and 2: score distributions and theoretical geometry

Read:

- Figure 1 shows the distributions of e^* , e^{**} , and e^{***} across subjects.
- Scores decline sharply when moving from basic rationalizability to FOSD-rationalizability.
- Figure 2 explains the probability-triangle geometry behind EUT and non-EUT theories.

Economic message: The empirical distribution and the theory diagram jointly show why it is not enough to focus on independence alone.

that all score have a median of one, the number of subjects who fail FOSD-rationalizability is about six times greater than those who are

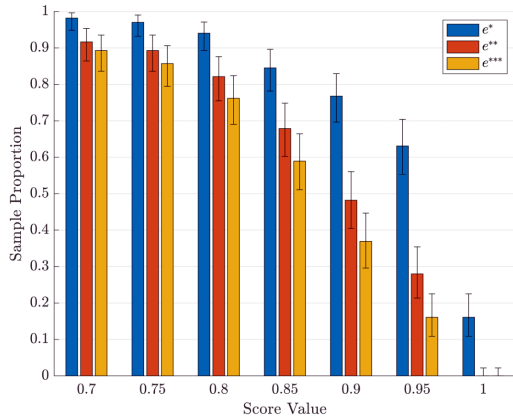


FIG. 1.—Distributions of rationalizability scores. The horizontal axis shows the score values, and the vertical axis presents the fraction of subjects with scores above each value for rationalizability (e^*), FOSD-rationalizability (e^{**}), and EUT-rationalizability (e^{***}). Whiskers extend to the minimum and maximum values.

Figure 1: rationalizability-score distributions

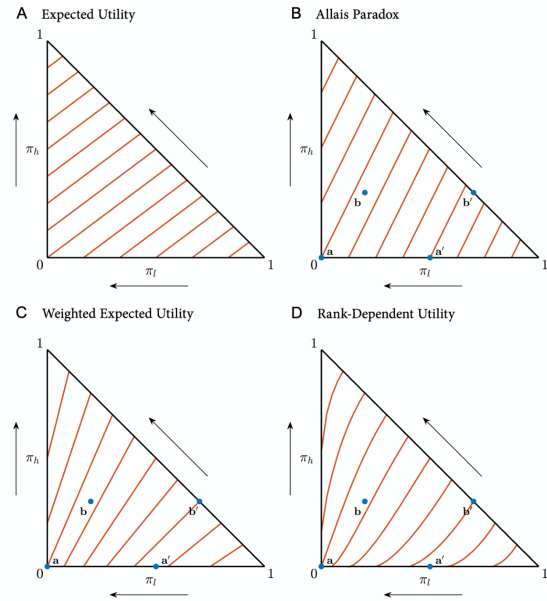


FIG. 2.—Indifference curves in probability triangle. The probability triangle depicts the lottery space as a set of probability weights (π_h, π_m, π_l) over three fixed outcomes (x_h, x_m, x_l) . Ordering (completeness and transitivity) plus continuity guarantee nonintersecting indifference curves; monotonicity (with respect to FOSD) guarantees that preferences are increasing as shown (arrows); adding independence gives rise to EUT, characterized by indifference curves that are parallel straight lines (A). The Allais paradox arises because EUT

Figure 2: probability-triangle models

6.2 Figures 3 and 4: experimental power and subject behavior

Read:

- Figure 3 shows that 3D budget data provide more power to reject EUT-rationalizability than 2D budget data.
- Figure 4 shows five individual subjects with different behavioral patterns, ranging from near EUT rationalizability to non-FOSD-rationalizability.

Economic message: The 3D design is both statistically more powerful and visually richer, revealing qualitatively different behavioral types.

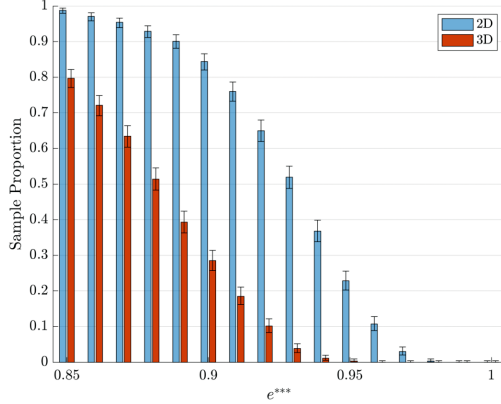


FIG. 3.—Power of testing EUT-rationalizability in 2D versus 3D experiments. Simulated objects whose choices are uniformly distributed on each budget set subject to perfect OSD-rationalizability ($e^{**} = 1$). Each simulated subject made 50 choices from randomly generated budget sets in the same way as the actual subjects. The horizontal axis shows the UT-rationalizability score values, and the vertical axis presents the fraction of simulated objects in the 2D and 3D experiments with scores above each value. Whiskers indicate

Figure 3: power in 2D versus 3D designs

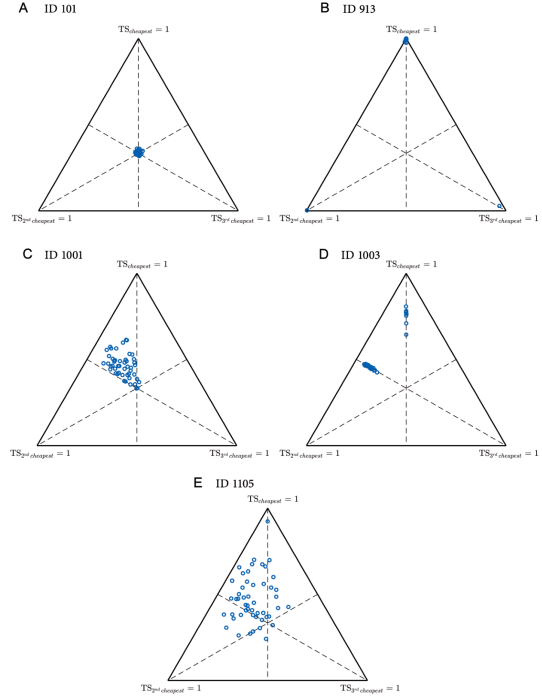


FIG. 4.—Subject behavior. Each plot shows all 50 choices for a single subject in terms of token shares. Each vertex of the unit simplex corresponds to a full allocation to one of the three securities. Some subjects are roughly EUT-rationalizable: ID 101 is consistent with in-

Figure 4: representative subject behavior

6.3 Table 1 and Figure 5: WARP/GARP and individual scores

Read:

- Table 1 reports the distribution of WARP and GARP violations and the CCEI scores needed to eliminate them.
- The median subject has 4 WARP violations and 7 GARP violations.
- Figure 5 plots individual rationalizability-score pairs.

Economic message: Basic consistency and monotonicity failures are quantitatively important. They cannot be dismissed as minor noise around EUT.

Percentile Value	Violations			CCEI Scores	
	WARP (1)	GARP (2)	GARP not WARP (3)	WARP (4)	GARP (5)
1	0	0	0	.668	.668
5	0	0	0	.787	.787
10	0	0	0	.828	.828
25	1	1	0	.916	.913
50	4	7	0	.969	.968
75	8	318	0	.996	.996
90	14	181,655	1	1.000	1.000
95	25	$\geq 10^7$	7	1.000	1.000
99	55	$\geq 10^7$	24	1.000	1.000

NOTE.—The table shows percentile values of the number of WARP violations, GARP violations, and GARP violations that do not contain a WARP violation along with the CCEI scores required to eliminate all WARP and GARP violations. Because of computational constraints, we are unable to fully compute the number of GARP violations that do not contain a WARP violation for three subjects (1.8%), each of whom has more than 10^7 GARP violations (even after removing WARP violations). For these subjects, we provide a lower bound on the number of GARP violations that do not contain a WARP violation.

Table 1: WARP/GARP violations and CCEI scores

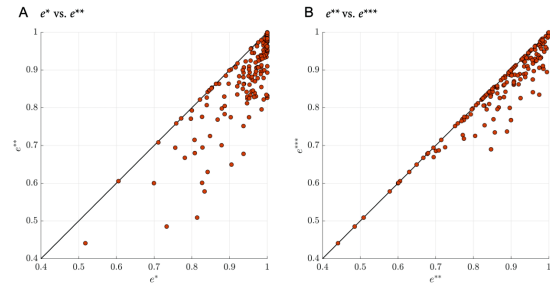


FIG. 5.—Individual-level rationalizability scores. A, e^* (horizontal axis) versus e^{**} (vertical axis). B, e^{**} (horizontal axis) versus e^{***} (vertical axis). By definition, $e^* \geq e^{**} \geq e^{***}$, so all points in both scatterplots must lie on or below the 45° line.

Figure 5: individual score scatterplots

6.4 Figures 6 and 7: main decomposition and subsample robustness

Read:

- Figure 6 compares the FOSD gap with the EUT gap for all subjects.
- Figure 7 compares the same gaps using 50 choices versus subsamples of 25 choices.
- The conclusion is robust: reducing the data generally reduces the measured FOSD gap but does not overturn the central pattern.

Economic message: The dominant source of EUT failure is not the independence axiom; the result survives a within-subject data reduction exercise.

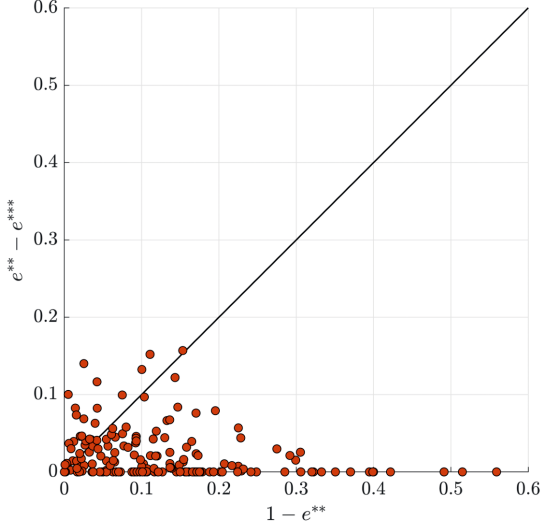


FIG. 6.—Difference-in-differences in individual-level rationalizability scores: difference between perfect rationalizability and FOSD-rationalizability ($1 - e^{**}$; horizontal axis) versus difference between FOSD-rationalizability and EUT-rationalizability ($e^{**} - e^{***}$; vertical axis).

between FOSD-rationalizability and EUT-rationalizability ($e^{**} - e^{***}$). Furthermore, for 125 subjects (74.4%), the difference between perfect rationalizability and FOSD-rationalizability ($1 - e^{**}$) is twice as large as

Figure 6: difference-in-differences decomposition

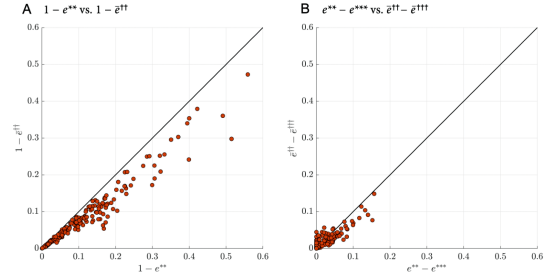


FIG. 7.—Individual-level differences in rationalizability scores (50 vs. 25 decisions). A, Difference between perfect rationalizability and FOSD-rationalizability ($1 - e^{**}$) with 50 decisions (horizontal axis) versus average difference ($1 - \bar{e}^{||}$) with 25 decisions (vertical axis). B, Difference between FOSD-rationalizability and EUT-rationalizability ($e^{**} - e^{***}$) with 50 decisions (horizontal axis) versus average difference ($\bar{e}^{||} - \bar{e}^{|||}$) with 25 decisions (vertical axis). By definition, $\bar{e}^{||} \geq e^{**}$ and $\bar{e}^{|||} \geq e^{***}$, so all points in A must lie on or below the 45° line, while points in B can lie below or above the 45° line.

Figure 7: 50 versus 25 decisions

6.5 Figures 8 and 9: statistical tests and benchmark comparisons

Read:

- Figure 8 reports statistical tests of the difference-in-differences decomposition at the individual level.
- Figure 9 compares actual subjects with simulated subjects whose choices satisfy FOSD but otherwise randomize.
- Actual subjects' gap between FOSD-rationalizability and EUT-rationalizability is smaller than the benchmark generated by random FOSD-rationalizable choices.

Economic message: Even relative to an FOSD-consistent random-choice benchmark, actual departures from EUT are mostly not of the kind emphasized by Allais-style independence violations.

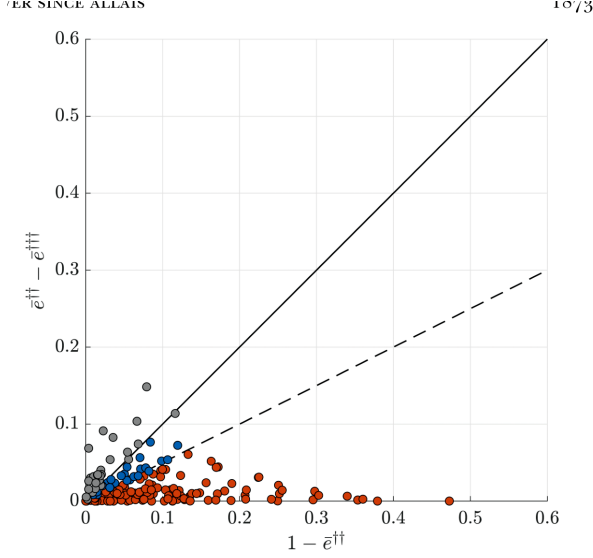


FIG. 8.—Statistical tests of difference-in-differences in individual-level rationalizability scores: average difference between perfect rationalizability and FOSD rationalizability ($1 - e^{II}$; horizontal axis) versus average difference between FOSD rationalizability and JT-rationalizability ($e^{III} - e^{II}$; vertical axis). The solid line represents the D_1 test, while the dashed line represents the D_2 test. Subjects are blue if only D_1 is positive and statistically significant, red if both D_1 and D_2 are positive and statistically significant, and gray if neither, at the 1% level.

th the gap between perfect rationalizability and FOSD-rationalizability

Figure 8: statistical tests

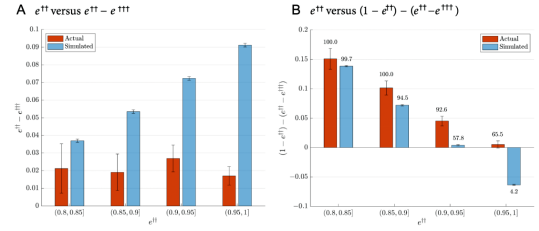


FIG. 9.—Departures from FOSD-rationalizability and EUT-rationalizability for actual versus simulated subjects. The horizontal axis shows different FOSD-rationalizability e^{II} score value bins. The vertical axis represents the mean difference between FOSD-rationalizability

Figure 9: actual versus simulated subjects

7 Identification and interpretation

7.1 What is identified by the design

- The design identifies whether each individual dataset satisfies revealed-preference restrictions.
- It identifies how much budget shrinkage is needed to rationalize the data under progressively stronger models.
- It identifies whether EUT failure mainly appears before or after imposing FOSD rationalizability.

Interpretation: The paper does not estimate preferences in the usual parametric sense. It measures the distance from observed behavior to classes of preferences.

7.2 Why individual-level analysis matters

- Pooled tests can hide heterogeneity across subjects.
- Individual-level CCEI scores allow the authors to classify each subject's source of inconsistency.
- Individual-level statistical tests avoid treating all subjects as if they share one preference model.

Interpretation: The paper's conclusion is stronger because it is not an aggregate artifact. The pattern appears for the vast majority of individuals.

7.3 Limitations

- Laboratory choices may not perfectly represent high-stakes field behavior.
- CCEI scores measure distance to rationalizability but do not explain psychologically why errors occur.

- The experiment is designed around three equally likely states, so extensions to unequal probabilities or more states may reveal additional patterns.
- Non-EUT models that also relax ordering or monotonicity are not the main comparison class.

Interpretation: The paper’s contribution is diagnostic rather than structural: it reveals which axioms fail, but not the full psychological process generating violations.

8 Short summary

- The paper provides a comprehensive experimental test of EUT axioms.
- It uses 3D budget sets with three state-contingent securities.
- The design separates WARP, GARP, FOSD, and EUT restrictions.
- The paper computes e^* , e^{**} , and e^{***} for every subject.
- Basic rationalizability is much more common than FOSD- or EUT-rationalizability.
- Violations of FOSD rationalizability are much larger than violations of EUT conditional on FOSD.
- The result weakens the empirical importance of Allais-style independence violations as the main explanation for EUT failure.
- Leading non-EUT models that retain ordering and monotonicity cannot explain most observed deviations from EUT.

9 Conclusion

9.1 Core conclusion

Dembo, Kariv, Polisson, and Quah show that the Allais tradition has focused too narrowly on independence. In their rich individual-level experimental data, most failures of EUT arise from failures of basic rationalizability and monotonicity with respect to FOSD, rather than from independence alone.

9.2 Economic intuition

A subject who violates EUT can do so for many reasons. She might have complete, transitive, monotone preferences but violate independence. Or she might make choices that are pairwise inconsistent, cyclically inconsistent, or dominated. The paper shows that the second set of failures is much more common.

9.3 Takeaway

The paper reframes the empirical evaluation of expected utility. It is not enough to compare EUT with non-EUT theories that merely relax independence. A complete account of risky choice must explain why many subjects violate more basic axioms such as ordering and monotonicity.